SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO2: *Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 40%

Duration: 1 Hour

QUESTIONS: 3 **Total Marks:30**

Annexure A

Dry Run Evaluation

Code of Conduct:

I K.Sasidhar Reddy(192525140) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

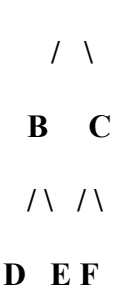
Signature of the student:*K.Sasidhar Reddy*

Annexure A

Dry Run Evaluation

1. Question 1 – Breadth-First Search (BFS)

Consider the following graph:



Task: Starting from node A, perform a Breadth-First Search (BFS). Complete the following table.

Iteration	Queue	Visited Node
1		
2		
3		
4		
5		
6		
7		

Questions:

1. **Write the BFS traversal order.**
2. **Show the queue contents after each iteration.**

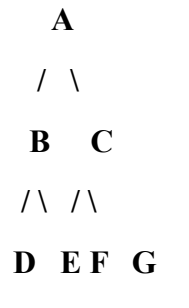
A) Graph: A → (B, C), B → (D, E), C → (F, G). Assume children are visited from left to right.

Iteration	Queue (front → rear)	Visited Node
1	[B, C]	A
2	[C, D, E]	B
3	[D, E, F, G]	C
4	[E, F, G]	D
5	[F, G]	E
6	[G]	F
7	[]	G

1. BFS traversal order: A, B, C, D, E, F, G.

2. Queue contents after each iteration are shown in the table. The queue is FIFO (First In, First Out).

2. **Depth-First Search (DFS) Using the same graph.**



Iteration	Stack	Visited Node
1		
2		
3		
4		
5		
6		
7		

Perform Depth-First Search (DFS) starting from A.

Complete the table.

A) Using left-to-right traversal, DFS explores each branch as deeply as possible before backtracking.

Step	Node visited	Stack (top at right; simplified)
1	A	[A]
2	B	[A, B]
3	D	[A, B, D]
4	E	[A, B, E] after backtracking
5	C	[A, C] after backtracking
6	F	[A, C, F]
7	G	[A, C, G]

DFS traversal order: A, B, D, E, C, F, G.

3)Initial State :

1 3 6
5 0 2
4 7 8

Goal State :

1 2 3
4 5 6
7 8 0

Task

Trace Breadth-First Search (BFS) until the goal state is reached.

Fill in the table.

Iteration Current State Queue Before Expansion Queue After Expansion
1
2
3

Iteration Current State Queue Before Expansion Queue After Expansion

4

5

Questions

1. **Which node is expanded in each iteration?**
2. **How many states are generated in total?**
3. **Write the complete solution path from the initial state to the goal state.**
4. **Why does BFS always find the shortest solution in an unweighted 8-puzzle problem?**

A) Initial state:

1 3 6
5 0 2
4 7 8

Goal state:

1 2 3
4 5 6
7 8 0

The five-row table in the question is not enough to reach the goal: the goal is reached at depth 8 (eight moves). The first five states removed from the BFS queue are listed below. Queue contents are abbreviated to avoid listing hundreds of states; BFS stores states in FIFO order.

Iteration	Current state expanded	Queue after expansion (summary)
1	1 3 6 / 5 0 2 / 4 7 8	New states from moving blank Up, Down, Left, Right (valid moves only)
2	1 3 6 / 5 2 0 / 4 7 8	Previously queued states remain first; newly generated states are appended
3	1 3 0 / 5 2 6 / 4 7 8	FIFO order maintained; repeated states are not added again
4	1 0 3 / 5 2 6 / 4 7 8	Unexpanded states remain in front of newly generated states
5	1 2 3 / 5 0 6 / 4 7 8	Successors are appended to the rear of the queue

1. Nodes expanded in the first five iterations are the five current states shown in the table.

2. The goal is first reached after expanding 311 states in this left/up/down/right neighbor ordering; 500 distinct states are discovered (including the initial state) before the goal

is dequeued. Counts can vary if the order of legal moves is changed.

3. One shortest solution path (blank movement): Right → Up → Left → Down → Left → Down → Right → Right.

Initial: 1 3 6 / 5 0 2 / 4 7 8

Step 1: 1 3 6 / 5 2 0 / 4 7 8

Step 2: 1 3 0 / 5 2 6 / 4 7 8

Step 3: 1 0 3 / 5 2 6 / 4 7 8

Step 4: 1 2 3 / 5 0 6 / 4 7 8

Step 5: 1 2 3 / 0 5 6 / 4 7 8

Step 6: 1 2 3 / 4 5 6 / 0 7 8

Step 7: 1 2 3 / 4 5 6 / 7 0 8

Goal: 1 2 3 / 4 5 6 / 7 8 0

4. BFS finds a shortest solution in an unweighted 8-puzzle because it explores states in increasing order of depth. Every move has equal cost, so the first goal state reached has the minimum number of moves.

Note: For the 8-puzzle, the original worksheet provides only five table rows, although reaching the goal requires more expansions. The full shortest path and explanation are included here.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Understanding of Search Problem	**20**	Clearly understands the search problem, initial state, goal state, and search strategy.	Understands the problem with minor misunderstandings.	Demonstrates partial understanding of the search problem.	Shows little or no understanding of the search problem.
State Expansion and Dry Run Execution	**30**	Correctly traces every iteration, expands nodes accurately, and maintains the Queue/Stack/Priority Queue without errors.	Performs most iterations correctly with minor mistakes in state expansion or data structure updates.	Performs only some iterations correctly; several errors in tracing or node expansion.	Unable to correctly perform the dry run or expand states.
Application of Search Algorithm	**20**	Applies the selected algorithm (BFS/DFS/UCS) correctly, following all algorithmic rules.	Applies the algorithm correctly with a few procedural errors.	Applies only basic algorithm steps with noticeable mistakes.	Fails to apply the correct search algorithm.
Accuracy of Solution Path	**20**	Identifies the correct traversal order/solution path and goal state with proper justification.	Solution path is mostly correct with minor errors.	Partially correct solution path; goal may not be reached correctly.	Incorrect or incomplete solution path.
Presentation and Logical Reasoning	**10**	Queue/Stack/Priority Queue updates, state diagrams, and explanations are neat, organized, and logically presented.	Presentation is clear with minor organizational issues.	Limited explanation and organization; reasoning is partially clear.	Poor presentation with unclear or missing reasoning.